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Karen Frilya Celine

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Education

National University of Singapore, PhD in Computer Science — Aug 2021 – present

- Thesis advisors: Warut Suksompong (primary), Frank Stephan, and Sanjay Jain

National University of Singapore, BSc (Hons.) in Applied Math — Aug 2016 – Aug 2021

- Thesis title: Bi-immunity for Functions
- Thesis advisors: Frank Stephan and Sanjay Jain
- CAP: 4.80/5.0

National University of Singapore, BComp (Hons.) in Computer Science — Aug 2016 – Aug 2021

- CAP: 4.71/5.0

Working Paper

1. Karen Frilya Celine and Warut Suksompong. Optimizing the envy cycle elimination algorithm. *CoRR*, abs/2606.02233, 2026

Journal Publications

1. Karen Frilya Celine, Warut Suksompong, and Sheung Man Yuen. On the fairness of additive welfarist rules. *ACM Transactions on Economics and Computation*, 14(2):5, Jun 2026
2. Karen Frilya Celine, Ziyuan Gao, Sanjay Jain, Ryan Lou, Frank Stephan, and Guohua Wu. Quasi-isometric reductions between infinite strings. *Journal of Computer and System Sciences*, 156:103718, Mar 2026
3. Cristian S. Calude, Karen Frilya Celine, Ziyuan Gao, Sanjay Jain, Ludwig Staiger, and Frank Stephan. Bi-immunity over different size alphabets. *Theoretical Computer Science*, 894:31–49, Nov 2021

Conference Publications

1. Karen Frilya Celine, Warut Suksompong, and Sheung Man Yuen. Comparing the fairness of recursively balanced picking sequences. In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 69–77, May 2026
2. Karen Frilya Celine, Warut Suksompong, and Sheung Man Yuen. On the fairness of additive welfarist rules. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 454–462, May 2025
3. Karen Frilya Celine. Balancing fairness and efficiency in the allocation of indivisible goods. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 2920–2922, May 2025. Doctoral Consortium track
4. Karen Frilya Celine, Ziyuan Gao, Sanjay Jain, Ryan Lou, Frank Stephan, and Guohua Wu. Quasi-isometric reductions between infinite strings. In *Proceedings of the 49th International Symposium on Mathematical Foundations of Computer Science (MFCS)*, pages 37:1–37:16, Aug 2024
5. Karen Frilya Celine, Muhammad Ayaz Dzulfikar, and Ivan Adrian Koswara. Egalitarian price of fairness for indivisible goods. In *Proceedings of the 20th Pacific Rim International Conference on Artificial Intelligence (PRICAI)*, pages 23–28, Nov 2023

Thesis

1. Karen Frilya Celine. Bi-immunity for functions. Bachelor's thesis, National University of Singapore, Singapore, Nov 2020

Teaching

Graduate Tutor, National University of Singapore — Aug 2021 – present

- CS1010S Programming Methodology: 2021/2022 Semester 2
- CS1231S Discrete Structures: 2021/2022 Semester 1, 2022/2023 Semester 2, 2023/2024 Semester 1, 2023/2024 Semester 2, 2024/2025 Semester 1, 2024/2025 Semester 2 (Head TA)
- CS3230 Design and Analysis of Algorithms: 2025/2026 Semester 2 (Head TA)
- CS4261/5461 Algorithmic Mechanism Design: 2024/2025 Semester 1

Undergraduate Teaching Assistant, National University of Singapore — Jan – Apr 2021

- CS1231 Discrete Structures: 2020/2021 Sem 2

Services

Program Committee Member

- Social Choice and Learning Algorithms (SCaLA) — 2026
- Games, Agents, and Incentives Workshop (GAIW) — 2026

Reviewer

- ACM Transactions on Economics and Computation — 2026
- Mathematics of Operations Research — 2025 – 2026
- International Joint Conference on Artificial Intelligence (IJCAI) — 2025

Presentations

On the Fairness of Additive Welfarist Rules

- 18th Meeting of the Society for Social Choice and Welfare (SSCW), Tokyo, Japan — Jun 2026
- 8th Games, Agents, and Incentives Workshop (GAIW), Paphos, Cyprus — May 2026
- 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Detroit, USA — May 2025

Comparing the Fairness of Recursively Balanced Picking Sequences

- 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Paphos, Cyprus — May 2026
- COMSOC Video Seminar Rump Session, Online — Apr 2026

Allocating Indivisible Goods: Fairness and Efficiency

- Doctoral Seminar, National University of Singapore, Singapore — Mar 2026

Balancing Fairness and Efficiency in the Allocation of Indivisible Goods

- 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Doctoral Consortium Track, Detroit, USA — May 2025

Egalitarian Price of Fairness for Indivisible Goods

- 20th Pacific Rim International Conference on Artificial Intelligence (PRICAI), Jakarta, Indonesia — Nov 2023

Internship

Software Engineering Intern, Google Singapore — May – Aug 2020

- Created a microapp to promote Group Buying, by leveraging on Google Pay's Spot Platform.
- Developed the application from scratch in a team of two, including designing the application's user flow, UI/UX mock, database schema and API endpoints, within 14 weeks.
- Responsible for full-stack development of Merchant-related features.

Awards and Scholarships

Research Achievement Award Aug 2025

- Awarded for outstanding research performance.

Full-Time Teaching Assistant Award Feb 2025

- Awarded to full-time teaching assistants with exceptional teaching performance.

ASEAN Undergraduate Scholarship Jul 2016 – Jun 2021

- Full scholarship awarded to high-achieving students from ASEAN countries.

Faculty of Science Dean's List Dec 2020

- Awarded to the top 5% of each cohort for academic performance in the preceding semester.

School of Computing Dean's List Dec 2020

- Awarded to the top 5% of each cohort for academic performance in the preceding semester.

Simon Marais Mathematics Competition Top Decile Individual Oct 2020

- Annual undergraduate mathematics competition across Asia Pacific region.

Dunearn Road Hostel/Sheares Hall Alumni Scholarship Sep 2018

- Awarded to residents of Sheares Hall who have done well academically while being actively involved in the hall for the preceding two academic years.

Skills

Programming languages: Python, JavaScript, Java, C/C++, HTML/CSS, SQL, LaTeX, MATLAB

Languages: English (native), Indonesian (native), Malay (proficient), Chinese (basic), Japanese (basic)